Department of Applied Mathematics and Statistics
COLORADO SCHOOL OF MINES
MATH 455: Partial Differential Equations

Consider the 2D curve $\Gamma = \{(x, \sin x) : -\frac{\pi}{2} \leq x \leq \frac{7\pi}{2}\}$. A point mass m slides along Γ under the acceleration g of gravity and the normal force (oriented up) of Γ on m. Whenever the normal force vanishes, m deviates from Γ and comes solely under the influence of g, until re-contacting the ramp. Assume restitution is zero. The point mass starts at the left endpoint of Γ and is imparted v_0 horizontal speed to the right. We are interested only in scenarios where v_0 is calibrated precisely so that m launches off the first hump of Γ, clears the trough centered at $x = 3\pi/2$, and has its first contact-after-launch on the down-ramp of Γ on $(5\pi/2, 7\pi/2)$. Find v_0 that minimizes the magnitude of m's incoming normal velocity at re-contact. Round your answer to 4 significant figures.

Solution — Let $x = a$ be the particle's launch point. We first work to find a as a function of v_0. While the particle is constrained to $y = \sin x$, conservation of energy determines the magnitude of its velocity at x by

$$v^2(x) = v_0^2 - 2g(\sin x + 1)$$

. By the standard unit normal formula for a parametrized curve, the upward unit normal at $(x, \sin x)$ is

$$\hat{\mathbf{n}} = \frac{(-\cos x, 1)}{\sqrt{1 + \cos^2 x}}$$

At any point x along the $y = \sin x$, the normal component of acceleration $a_n = v^2\kappa$ (where κ is 2D curvature) is

$$\frac{-v^2 \sin x}{(1 + \cos^2 x)^{3/2}}$$

(This is the normal acceleration required to follow the curve). The gravitational component of acceleration perpendicular to the ramp is

$$a_g = (0, -g) \cdot \hat{\mathbf{n}} = \frac{-g}{\sqrt{1 + \cos^2 x}}$$

The gravitational normal acceleration and the normal acceleration supplied by the ramp's normal force must sum to the net normal acceleration required to follow the curve:

$$\frac{N}{m} + \frac{-g}{\sqrt{1 + \cos^2 x}} = -\frac{v^2 \sin x}{(1 + \cos^2 x)^{3/2}}$$

At launch $x = a$, the normal force vanishes ($N = 0$), giving:

$$v_a^2 = g\left(\frac{1 + \cos^2 a}{\sin a}\right)$$

where v_a is speed at loss of contact. From conservation of energy, we obtain

$$v_0^2 - v_a^2 = 2g(\sin a + 1)$$

$$v_0^2 - g\left(\frac{1+\cos^2 a}{\sin a}\right) = 2g(\sin a + 1)$$

$$\implies v_0 = \sqrt{2g(\sin a + 1) + g\left(\frac{1+\cos^2 a}{\sin a}\right)}$$

We now have v_0 as a function of the particle's launch point a, as we intended. Now we proceed to find a (and by extension v_0) such that the particle has an optimal landing back on the track. Note that the unit tangent at $x = a$ is

$$\frac{(1, \cos a)}{\sqrt{1+\cos^2 a}}$$

so the particle's components of velocity here simplify to

$$v_x = \sqrt{\frac{g}{\sin a}} \qquad v_y = \cos a\sqrt{\frac{g}{\sin a}}$$

Let $x = b$ be the particle's touch-down point. Since horizontal velocity is constant while airborne, the air time must be

$$t_{air} = (b-a)\sqrt{\frac{\sin a}{g}}$$

and (by basic projectile motion), its height at b must be

$$y_b = \sin a + (b-a)\cos a - \frac{1}{2}\sin a(b-a)^2$$

Recontact with the ramp requires $\sin b = y_b$. Let us now obtain the incoming normal velocity. Right before contact, we have $v_x = \sqrt{g/\sin a}$, and

$$v_y = \cos a\sqrt{\frac{g}{\sin a}} - gt_{air} = \sqrt{\frac{g}{\sin a}}(\cos a - (b-a)\sin a)$$

The normal component of velocity at contact is

$$v_n = (v_x, v_y)\cdot \hat{\mathbf{n_b}} = (v_x, v_y)\cdot\frac{(-\cos b, 1)}{\sqrt{1+\cos^2 b}}$$

$$= \frac{\sqrt{g/\sin a}}{\sqrt{1+\cos^2 b}}(\cos a - \sin a(b-a) - \cos b)$$

The minimum possible value of $|v_n|$ is zero, so we see if a launch point a exists that forces $v_n = 0$ at re-contact. $v_n = 0$ is satisfied when

$$\cos a - \sin a(b-a) = \cos b$$

From this equation and from the constraint $y_b = \sin b$, we have a system of two equations in $\sin a$ and $\cos a$:

$$\begin{cases} \cos a - \sin a(b-a) = \cos b \\ \sin a + (b-a)\cos a - \frac{1}{2}\sin a(b-a)^2 = \sin b \end{cases}$$

(Note that the first equation enforces tangency and the second equation enforces contact). Let's simplify the notation with $d = b - a$:

$$\begin{cases} \cos a - \sin a(d) = \cos(a+d) \\ \sin a + d\cos a - \frac{1}{2}\sin a(d)^2 = \sin(a+d) \end{cases}$$

and expand $\cos(a+d)$ and $\sin(a+d)$ to express the system as a linear, homogeneous system in $(\sin a, \cos a)$:

$$\begin{cases} \cos a(1-\cos d) + \sin a(\sin d - d) = 0 \\ \cos a(d - \sin d) + \sin a(1 - \frac{1}{2}d^2 - \cos d) = 0 \end{cases}$$

Since $\sin a$ and $\cos a$ can't both be zero, this system can only be satisfied when it has a non-trivial solution: equivalently, when its determinant is zero:

$$(1-\cos d)(1 - \frac{1}{2}d^2 - \cos d) + (d - \sin d)^2 = 0$$

$$\implies d = 8.98682$$

$$\implies a = 0.21898$$

$$\implies b = a + b = 9.2058 \quad \text{(landing position)}$$

The particle's landing position is admissible because it lies in the track's down-ramp interval $(5\pi/2, 7\pi/2) = (7.85, 10.99)$ In order for the particle to lift off at our found value of $x = a$, our function $v_0(a)$ dictates that $v_0 = 10.5808384338$. We round to 4 significant figures:

$$\boxed{v_0 = 10.58}$$